

Comprehensive Vehicle Development of MY2029 Honda Civic

1. Introduction

This project presents a comprehensive end-to-end vehicle development study for a MY2029 Honda Civic in the compact sedan segment. The objective is to develop a future-ready vehicle concept by integrating market analysis, engineering design, system-level modeling, and simulation-based validation.

The study focuses on a mild hybrid vehicle architecture aimed at improving fuel efficiency, emissions performance, and overall vehicle competitiveness.

2. Market Analysis and Benchmarking

A detailed benchmarking study was conducted using key competitors in the compact sedan segment including Honda Civic, Toyota Corolla, Hyundai Elantra, Mazda3, Nissan Sentra, and Volkswagen Jetta.

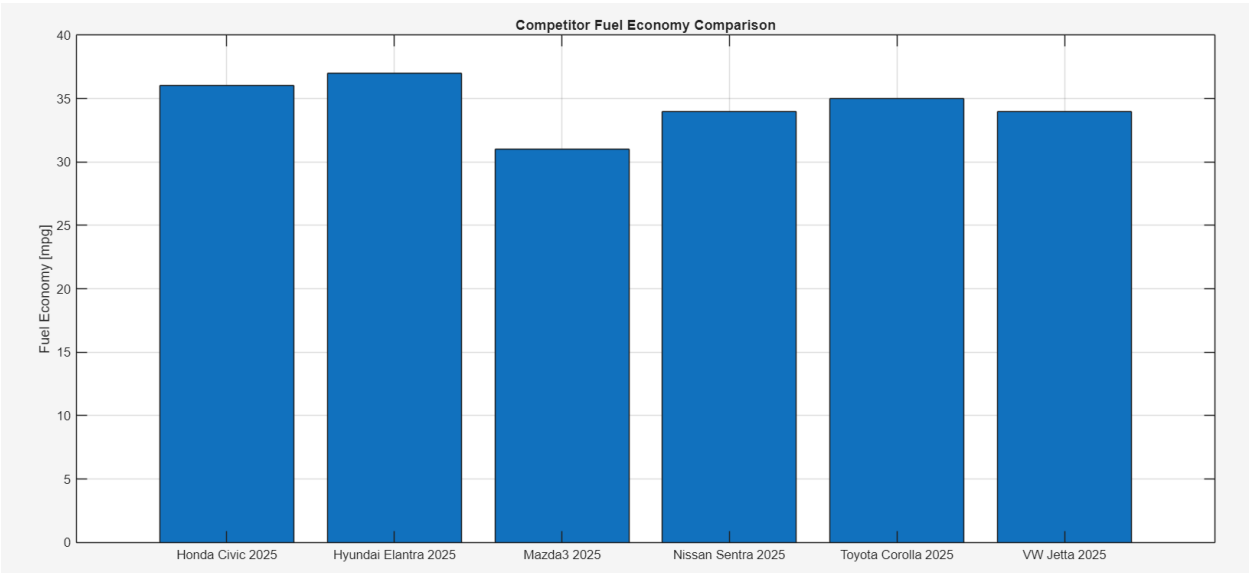
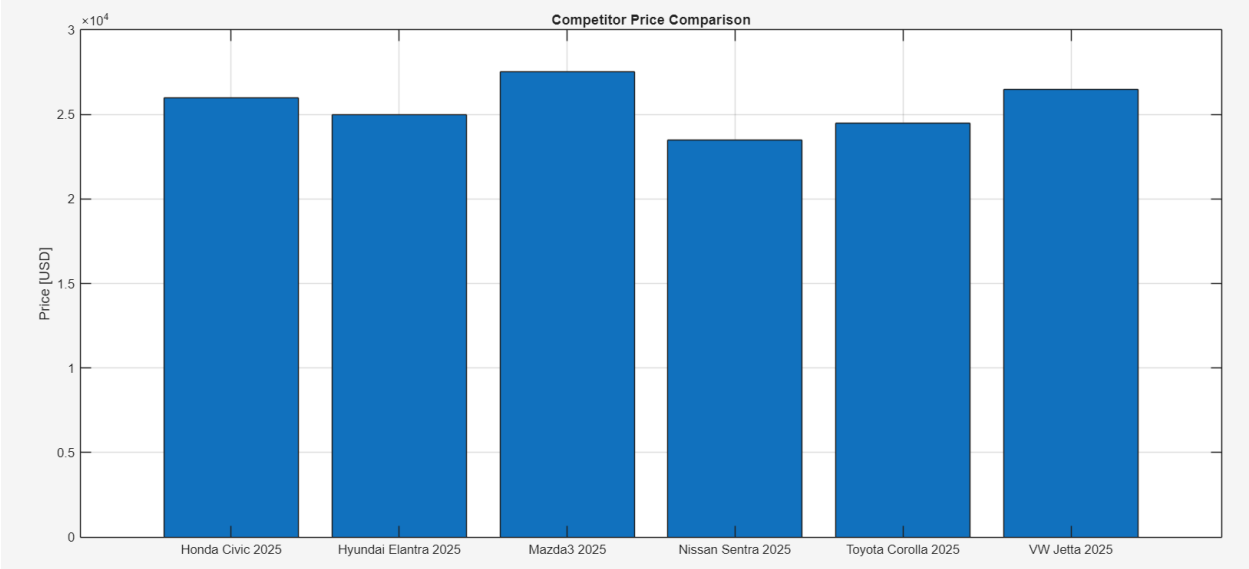
The benchmarking analysis considered:

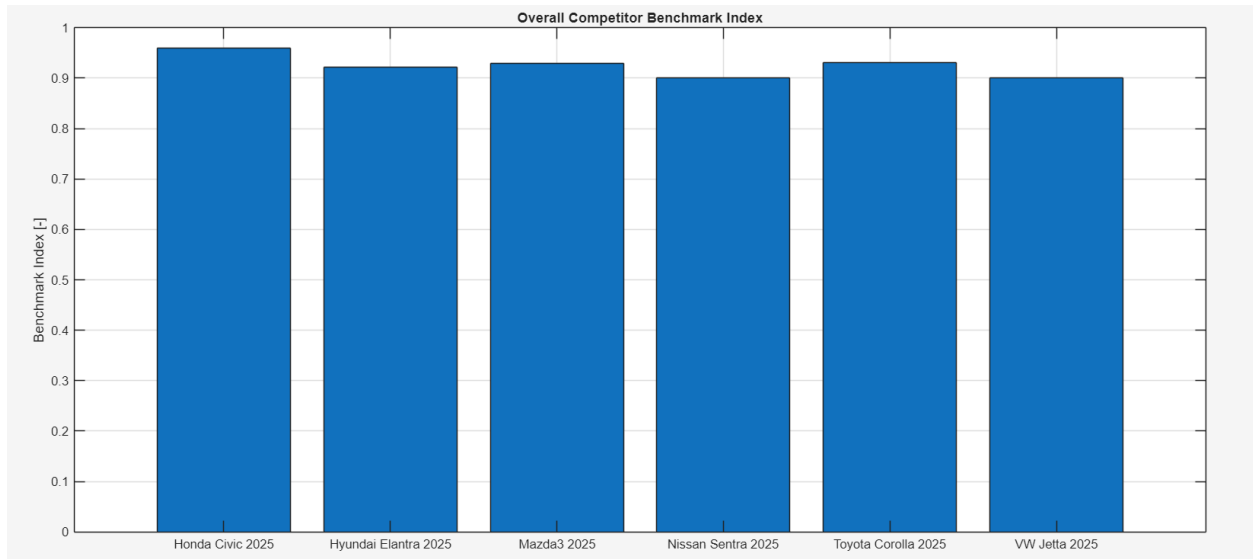
- Price
- Power
- Fuel economy
- Vehicle mass
- Acceleration performance
- Cargo capacity
- Safety, technology, and comfort scores

Results indicate that:

- The Honda Civic 2025 achieved the highest benchmark index (0.96027)
- The average market price is approximately \$25,500
- The average fuel economy is 34.5 mpg

This analysis establishes a competitive baseline for the MY2029 Civic development.





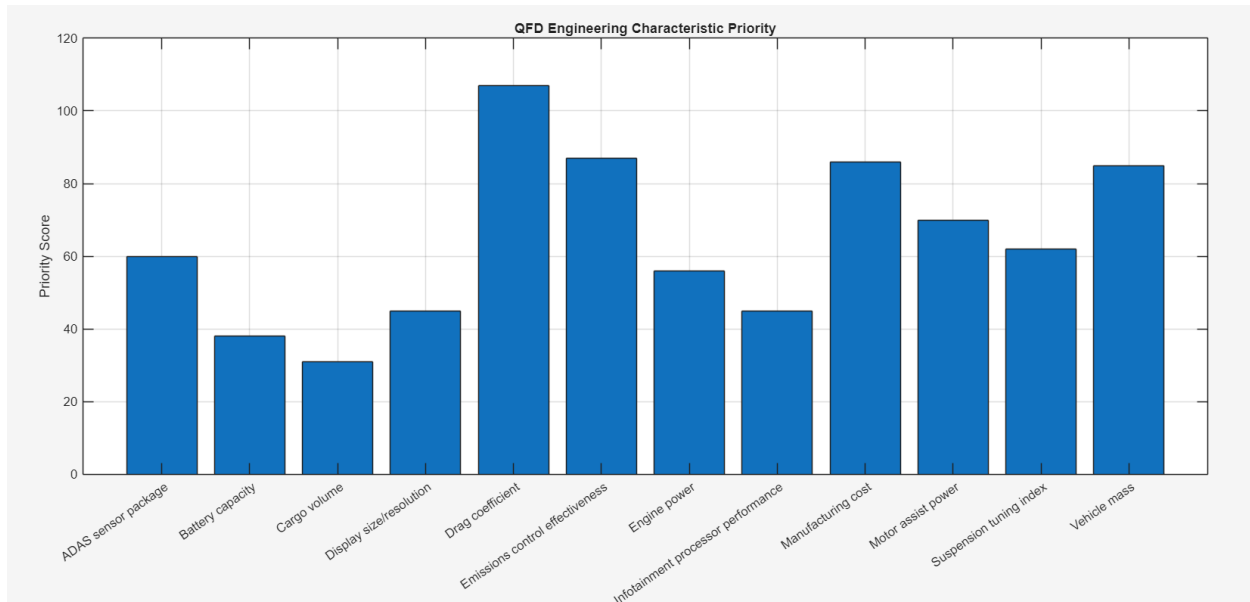
3. Customer Needs and QFD Analysis

A Quality Function Deployment (QFD) approach was used to translate customer needs into engineering characteristics.

Key findings:

- **Drag coefficient (13.86%)** is the highest priority engineering parameter
- Emissions control and manufacturing cost are also critical
- Vehicle mass and hybrid motor assist play significant roles

This ensures that engineering efforts are aligned with customer expectations.



4. Target Vehicle Specifications

Based on benchmarking and QFD analysis, the following targets were defined:

- Target power: 190 hp
- Fuel economy: 45 mpg
- Mass: 1425 kg
- 0–60 mph: 7.3 seconds
- Drag coefficient: 0.25
- Battery capacity: 1.8 kWh
- Motor assist: 18 kW

The vehicle adopts a **front-wheel-drive mild hybrid architecture** with a focus on ULEV emissions compliance.

5. Requirements Cascade and Traceability

A structured requirements cascade was developed:

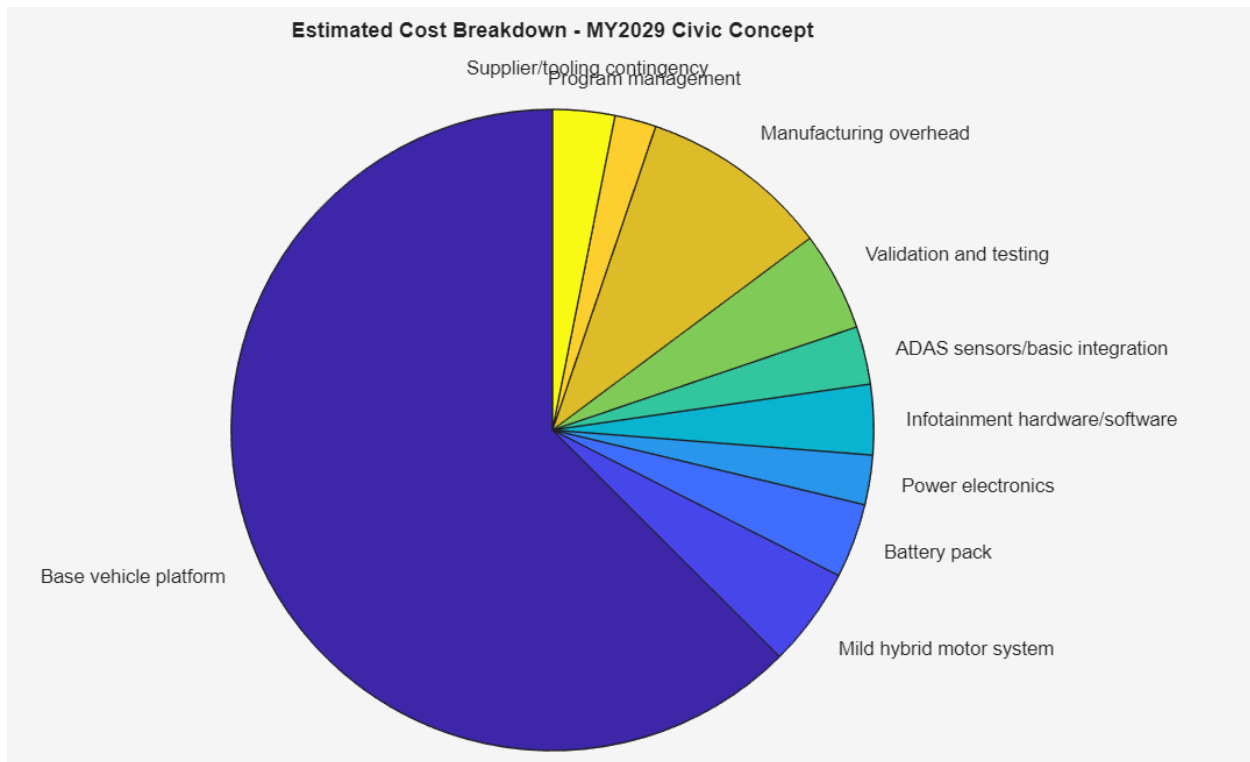
- Customer Requirements (CR)
- Engineering Requirements (ER)

- Subsystem Requirements (SR)

Each requirement is linked with:

- Parent requirement
- Verification method (Simulation, Testing, MIL/SIL/HIL)

This ensures full traceability and validation readiness.



6. Vehicle Sizing Analysis

Vehicle sizing calculations were performed to estimate power and force requirements.

Key results:

- Required powertrain power: **~220 hp**
- Total force at 60 mph: **5646 N**
- Wheel power requirement: **151.44 kW**

These results validate that the selected powertrain meets performance targets.

7. Financial Development Plan

A cost model was developed including:

- Base vehicle platform
- Hybrid system components
- Battery and electronics
- ADAS and infotainment systems

Total estimated variable cost:

\$24,000 per vehicle

This supports feasibility within the target price range.

8. Technology Roadmap

A phased development plan was created:

- **2026:** Concept and benchmarking
- **2027:** Subsystem and hybrid development
- **2028:** Simulation and validation
- **2029:** Final validation and launch

This roadmap aligns with industry-standard OEM development cycles.

9. Vehicle Validation Plan

A comprehensive validation framework was defined including:

- Fuel economy simulation
- Acceleration testing
- SOC behavior
- Infotainment performance
- ADAS functionality

This ensures readiness for production validation.

10. Simulink-Based Vehicle Model

A longitudinal vehicle model was developed in Simulink, including:

- PI speed controller
- Engine-motor torque split
- Aerodynamic and rolling resistance
- Battery SOC model

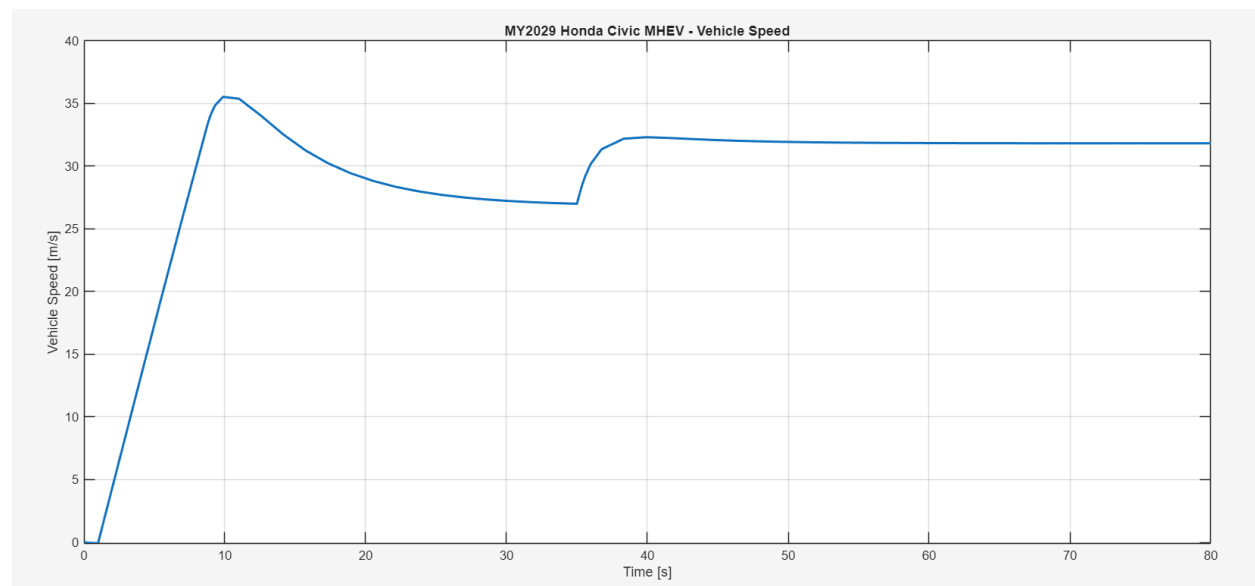
This model represents the functional behavior of the vehicle.

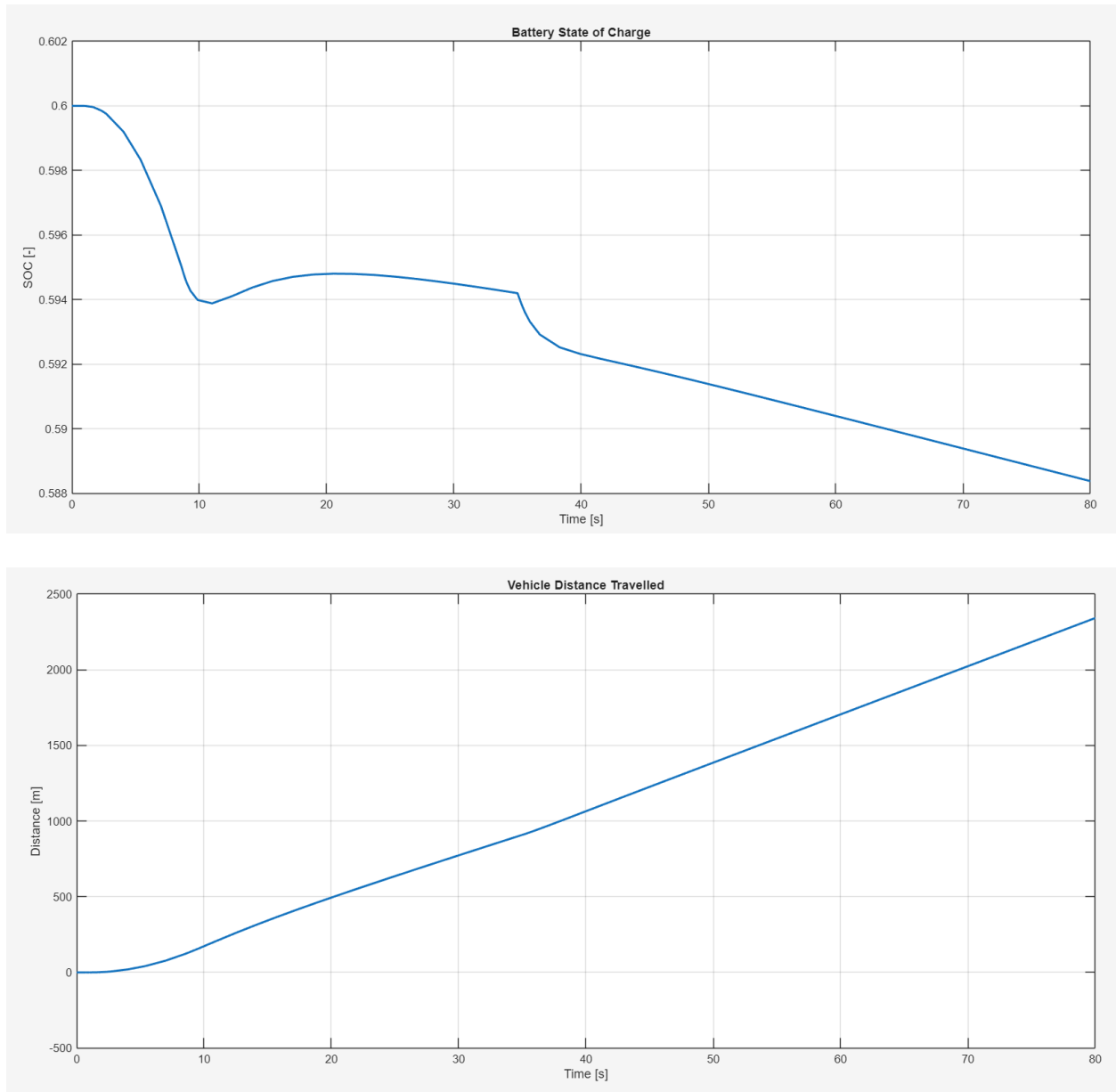
11. Simulation Results

Simulation results show:

- Maximum speed: **79.47 mph**
- Estimated 0–60 mph: **7.59 seconds**
- Total distance covered: **2341 m**

The simulation closely matches the target acceleration (7.3 s), validating the design.





12. Conclusions

The MY2029 Honda Civic concept demonstrates:

- Strong alignment with market expectations
- Improved fuel efficiency and emissions performance
- Competitive acceleration and power
- Feasible cost structure

The integration of MATLAB and Simulink enables early-stage validation, reducing development risk.